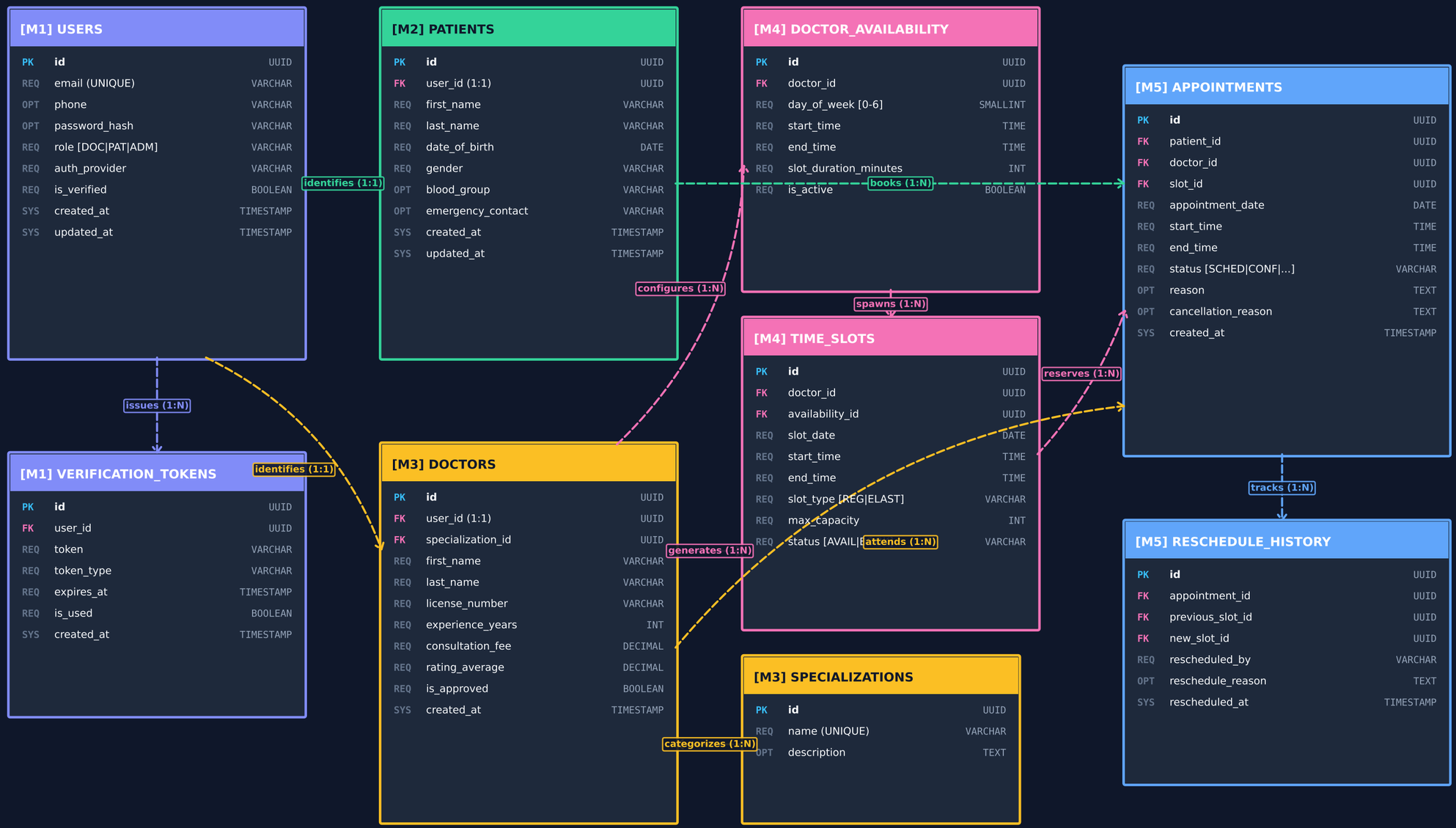
SCHEDULA • HOSPITAL MANAGEMENT SYSTEM

Schedule Project Module • Entity-Relationship (ER) Architecture (5 Modules)

1. User Module | 2. Patient Module | 3. Doctor Module | 4. Availability Module | 5. Appointment Module



SCHEDULA • DATA DICTIONARY & ARCHITECTURAL SPECIFICATIONS

Schedule Project Module • Relational Constraints & 3NF Normalization

Module	Table Name	Key Columns	Data Types	Constraints & Foreign Keys	Description
1. User	users	id, email, password_hash, role, auth_provider	UUID, VARCHAR(255), VARCHAR(20)	PK, UNIQUE(email), NOT NULL role IN ('DOCTOR','PATIENT','ADMIN')	Central authentication authority; decoupled from clinical profiles
1. User	verification_tokens	id, user_id, token, expires_at, is_used	UUID, VARCHAR(255), TIMESTAMPTZ, BOOL	PK, FK -> users(id) ON DELETE CASCADE	Secure token management for OTP, Google OAuth, and email verification
2. Patient	patients	id, user_id, first_name, last_name, dob, blood_group	UUID, VARCHAR(100), DATE, VARCHAR(5)	PK, FK -> users(id) UNIQUE (1:1) ON DELETE CASCADE	Patient demographic identity and baseline medical profile attributes
3. Doctor	specializations	id, name, description	UUID, VARCHAR(100), TEXT	PK, UNIQUE(name), NOT NULL	Standardized medical specialties (Cardiology, Pediatrics, General)
3. Doctor	doctors	id, user_id, specialization_id, license_number, fee, ratings	UUID, VARCHAR(50), DECIMAL(10,2), INT	PK, FK -> users(id) UNIQUE (1:1) FK -> specializations(id)	Doctor credentials, medical license, consultation charges & admin approval
4. Availability	doctor_availability	id, doctor_id, day_of_week, start_time, end_time, duration	UUID, SMALLINT, TIME, INT	PK, FK -> doctors(id) ON DELETE CASCADE	Weekly recurring schedule rules (e.g., Mon-Fri 09:00 - 17:00, 30m slots)
4. Availability	time_slots	id, doctor_id, slot_date, start_time, max_capacity, type	UUID, DATE, TIME, VARCHAR(30), INT	PK, FK -> doctors(id) slot_type: REGULAR | ELASTIC	Concrete booking calendar slots; supports Week 3 elastic capacity
5. Appointment	appointments	id, patient_id, doctor_id, slot_id, status, reason	UUID, DATE, TIME, VARCHAR(30), TEXT	PK, FK -> patients, doctors, slots status: SCHED|CONF|COMPL|CANC	Core booking transaction linking patient, doctor, and reserved calendar slot
5. Appointment	reschedule_history	id, appointment_id, prev_slot_id, new_slot_id, reason	UUID, VARCHAR(20), TEXT, TIMESTAMPTZ	PK, FK -> appointments(id) ON DELETE CASCADE	Full historical audit trail of rescheduled slots with reasons